CRITICAL PREFACE

Germany as an Industrial Hub for a New Europe

Why this proposal is retained but not endorsed

This preface records a critical reassessment of the proposal that follows. The original publication is reproduced in full and without substantive edits. It is preserved as a record of the analysis developed at the time, not as a current recommendation for implementation. The reassessment below explains why I consider the proposal unsuccessful in its central political-economic design.

The fundamental problem: security can become an economic growth imperative

The original proposal treated public procurement—including defence procurement—as one possible way to create initial demand, scale production, and accelerate the development of technologies that might later find civilian markets. In engineering and industrial terms, that mechanism can be plausible. The difficulty is that defence demand is not economically or politically neutral. If it becomes a principal engine of industrial renewal, the livelihoods of firms, workers, investors, and regions can become increasingly dependent on continuing or expanding military orders.

This creates a structural conflict of incentives. Actors may come to have a material interest in larger defence budgets, sustained threat perceptions, and the continued expansion of arms production—not necessarily because they desire war, but because their income, employment, investment returns, and local economic stability become tied to that trajectory. The question then extends beyond European security. It also concerns the financial well-being of ordinary households whose wages, jobs, taxes, and public services are affected by how economic resources are allocated.

The security dilemma and the risk of escalation

A state may expand its military capacity in order to feel safer. Other states, unable to know with certainty whether that capacity is defensive or may later be used against them, can respond by expanding their own. Each side can interpret its actions as prudent self-protection while interpreting the other side’s response as evidence of growing danger. The resulting feedback loop can make all parties less secure, even when no participant initially seeks a war.

This is a security-dilemma dynamic and can, under some conditions, contribute to the kind of self-reinforcing rivalry associated with the Thucydides Trap. The analogy is a warning, not a claim that war is inevitable or that the present situation is identical to the period before the Second World War. Historical circumstances differ. The relevant lesson is that the interaction of threat perceptions, arms competition, political commitments, and entrenched economic interests can narrow the room for de-escalation over time.

External beneficiaries and the difficulty of changing course

A sustained European defence build-up could also create opportunities for external powers and suppliers—including the United States and China in different parts of the industrial and technology landscape—to benefit from sales, investment, components, or the wider reorganisation of production. The extent of such benefits would depend on actual policy choices and supply chains; it should not be assumed in advance. The structural risk is that once substantial capital, employment, political influence, and cross-border commercial relationships depend on a high level of military demand, changing direction may become more difficult. Economic stakeholders can reinforce a policy path even when its broader security consequences are harmful.

Why the proposal is closed as a strategic case

The proposal did not establish sufficient safeguards to ensure that defence procurement would remain a limited, subordinate instrument rather than becoming a central driver of industrial policy. Its suggested transition from initial public demand to civilian markets is possible in principle, but it is not automatic; nor does a five-year review or an exit condition by itself resolve the incentives created by a large and durable defence economy. The design therefore leaves a fundamental problem unresolved: a programme intended to increase security and prosperity could help create constituencies whose material interests favour continued militarisation, provoking counter-militarisation and increasing the risk of conflict.

For that reason, I am closing this proposal as an unsuccessful strategic case rather than recommending it for implementation. This is not a conclusion that industrial renewal, automation, affordable energy, or European economic cooperation are undesirable. It is a conclusion that the architecture presented in the original paper does not adequately separate civilian industrial development from the self-reinforcing incentives of an arms race. Replacing its central pillar with the Mittelstand would not, on its own, solve this problem; nor is there sufficient basis to claim that one category of company is inherently better suited to provide the answer.

Scope of the retained original

The publication that follows remains unchanged, including its original argument, recommendations, and references to defence procurement and dual-use development. Those passages should be read in light of this preface. Their inclusion documents the proposal as it was originally formulated; it does not mean that I continue to endorse its overall architecture.

Author’s note — This preface is a retrospective assessment of the proposal’s strategic logic. It identifies a structural risk, not a prediction that war is inevitable, and does not claim that every defence contract or military capability necessarily produces escalation.

ORIGINAL PROPOSAL

(Reproduced without substantive edits)

Germany as an Industrial Hub for a New Europe

A Political-Economic Initiative for the Transition to a New Production Model

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Prepared with analytical assistance from ChatGPT and Grok.

Objective

The objective is not to preserve Germany's existing industrial structure at any cost, but to transform its productive base before industrial contraction deprives the country of the resources needed for independent development.

The proposal is to integrate competitive energy, artificial intelligence, computing capacity, robotics, new high-tech markets, and public-sector demand into a coherent industrial renewal programme.

At the same time, two interconnected objectives must be achieved:

1. Preserve Germany's ability to independently develop, scale, and export complex physical systems.
2. Make the new production platform a foundation for European industrial development, avoiding both Germany's isolation within the EU and the loss of its independent role in shaping Europe's economic architecture.

The central principle is:

> *Germany must use its existing industrial and political weight to transition to a new production model while it still has sufficient weight to influence the architecture of that transition.*

1. Diagnosis: Why the Existing Model Is Losing Its Effectiveness

Germany's post-war industrial model relied on a combination of advantages:

- Affordable and relatively reliable energy.
- A strong engineering tradition.
- Advanced industrial equipment and technological capabilities.
- Dense supplier networks.
- A skilled workforce.
- Access to major international markets.
- Exports of complex, high-value-added products.

These advantages reinforced one another. However, the external environment is changing, and the continuation of the conditions that supported this model can no longer be taken for granted.

1.1. Energy: From Competitive Advantage to Constraint

For energy-intensive industries, the price and availability of energy directly affect production costs.

The causal chain is straightforward:

> *Expensive energy → higher production costs → declining price competitiveness → lower margins → reduced investment → relocation or closure of production facilities → erosion of the industrial base.*

The consequences extend beyond individual businesses:

> *Declining production → weaker export capacity and industrial employment → erosion of the tax base and supplier networks → fewer resources available for future investment.*

If this process becomes persistent, Germany loses not only individual production facilities but also its ability to reproduce its own industrial capabilities.

1.2. China: A Changing Competitive Environment

China can no longer be viewed exclusively as a source of inexpensive products with lower technological sophistication.

In several industries, Chinese manufacturers can already compete with European producers in quality, functionality, and technological characteristics. At the same time, they may benefit from economies of scale, integrated supply chains, resource advantages, and lower production costs.

The causal chain is:

> *Technological convergence → continued Chinese cost advantages → displacement of German products in selected markets → declining production volumes → weaker incentives for investment and R&D → reduced capacity to develop the next generation of competitive products.*

This creates a risk of self-reinforcing industrial decline.

The challenge is no longer simply to produce existing goods more cheaply than competitors. Germany must transition to a production model that enables it to develop new products and scale them at competitive unit costs.

1.3. The New Production Function: Affordable Energy, Robots, and Computing

The combination of more affordable energy, robotics, and computing capacity could fundamentally change the economics of industrial production.

As the cost of human labour becomes less decisive for an increasing number of operations, advantages shift towards manufacturers capable of:

- Automating production lines.
- Applying industrial AI.
- Integrating robotics and machine vision.
- Managing production through data-driven systems.
- Automating logistics and quality control.
- Rapidly reconfiguring production processes.
- Scaling output while reducing marginal costs.

Germany risks becoming trapped between two production models: its existing industrial structure becomes too expensive, while the new model has not yet been established at sufficient scale.

If other countries transition to automated production earlier, Germany could lose its position not only in traditional industries but also in emerging production networks.

The central threat is not simply the loss of individual markets, but the loss of the ability to scale complex physical production at competitive costs.

1.4. The Self-Reinforcing Cycle of Industrial Decline

Taken together, these factors could trigger the following sequence:

> *Expensive energy → uncompetitive products → lost markets → capital outflows → factory closures → loss of suppliers and expertise → declining R&D → reduced technological capabilities → further loss of competitiveness.*

At a certain point, recovery becomes substantially more difficult. Restarting an individual business can be faster than rebuilding an entire industrial ecosystem, including specialists, suppliers, equipment, engineering knowledge, and reliable demand.

Industrial capabilities also deteriorate nonlinearly. Once a critical mass of suppliers, engineers, and serial-production experience is lost, rebuilding the ecosystem may become not merely expensive but practically impossible at its previous scale within a single political cycle.

The consequences could affect:

- Exports.
- Labour productivity.
- The tax base.
- Skilled employment.
- Defence-industrial capacity.
- Technological autonomy.
- Germany’s bargaining power within Europe.

Industrial renewal is therefore not merely a matter of economic policy. It is directly connected to the country’s long-term ability to determine its own development trajectory.

2. What Must Actually Be Preserved

Government policy should not assume that every existing business, product, or job must be preserved in its current form.

That confuses two different objectives:

1. Preserving the historically established structure of industry.
2. Preserving the country’s ability to turn engineering knowledge, capital, and technology into competitive physical products.

The second objective is more important.

Germany must preserve and develop:

- Engineering capabilities.
- Production technologies.

- The ability to design complex systems.
- Industrial equipment and machinery.
- Skilled technical personnel.
- Critical supplier networks.
- Research and development capabilities.
- The ability to move rapidly from development to serial production.

Germany’s Mittelstand deserves particular attention: the extensive network of specialised small and medium-sized companies that can be global leaders in narrowly defined technological segments.

Historically, this network has provided depth to German supply chains and enabled the country to move from development to serial production across a wide range of industrial applications.

In the new production model, the Mittelstand should not merely be an object of protection. It should become one of the principal carriers of automation, industrial AI, and new production platforms.

This would allow existing industries to transform rather than simply preserve their former positions.

For example, the automotive industry may undergo substantial changes in its products and production models while retaining some of its engineering capabilities, equipment, suppliers, and manufacturing capacity.

The same principle applies to machinery, energy equipment, industrial automation, and other sectors.

The state’s task is not to freeze the existing industrial structure, but to enable the industrial base to transition to the next generation of production.

3. The Proposed Solution: A New Production Architecture

The programme should integrate four areas of transformation.

3.1. Competitive Energy

Germany needs an energy system that provides industry with predictable costs, reliable supply, and the ability to plan investments over the long term.

Nuclear power should be reconsidered as one potential component of this system.

This does not mean excluding all other sources. On the contrary, resilience should come from combining complementary elements:

- Nuclear generation.
- Renewable energy.
- Energy storage.
- Gas-fired reserve generation.
- LNG infrastructure.
- Electricity grids and interconnectors.
- Demand management and peak-load balancing.

The specific configuration should be determined by considering full lifecycle costs, deployment timelines, reliability, grid constraints, and industrial requirements.

The selection criterion should not be whether a technology belongs to a particular political programme, but whether it contributes to the competitiveness and resilience of the energy system.

The principle of multi-layered energy security

Existing gas infrastructure should not automatically be considered redundant simply because long-term priorities are changing.

It may retain value as a reserve, a safeguard against supply disruptions, a balancing instrument, and a means of strengthening resilience during the transition.

The state should evaluate this infrastructure not only by its current utilisation rate but also by the cost of maintaining the readiness it provides.

In other words, some expenditure may represent a deliberate payment for energy optionality: the ability to use reserve capacity when circumstances deteriorate.

At the same time, past investment must not become an argument for blocking a more efficient future energy system merely because capital has already been spent.

Existing infrastructure should preserve freedom of transition, not prevent it.

3.2. Transition to Automated Production

An improved energy base alone will not restore competitiveness if production processes remain expensive and slow.

Germany must systematically develop:

- Industrial artificial intelligence.
- Robotic production lines.
- Machine vision.
- Automated process control.
- Digital twins.
- Predictive maintenance.
- Autonomous intralogistics.
- Automated quality control.
- Integration of product design, manufacturing, and operation.

The objective is not simply to introduce individual technologies, but to change the economics of production.

> *Energy → computing → AI → robotics → automated production → higher productivity → lower unit costs.*

Automation will not eliminate all labour costs or automatically make every product competitive. Its results will depend on production structure, capital expenditure, equipment costs, component availability, and the ability to scale.

However, the systematic integration of these technologies creates an opportunity to compete through productivity rather than relying on continuous reductions in wages.

3.3. Developing New High-Tech Markets

Automating existing production is necessary but insufficient. Germany also needs new products and markets that can use its accumulated engineering capabilities and generate additional demand.

Candidate areas for assessment include:

- Industrial AI and robotics.
- Autonomous systems.
- Intelligent manufacturing equipment.
- Energy management systems.
- High-end machinery.
- Industrial inspection and sensing systems.
- Dual-use platforms.
- Aerospace and advanced sensing.
- Medical and pharmaceutical technologies.
- Intelligent infrastructure.

Not all of these markets should be financed simultaneously. Each direction must be assessed for Germany’s real advantages, potential demand size, technological entry barriers, export potential, and the ability to reach serial production.

Key criteria:

1. Use of existing engineering and production competencies.
2. Potential for substantial productivity gains through automation.
3. Sufficient potential market.
4. Ability to create technological advantages that are difficult to reproduce quickly.
5. Capacity to transition from public support to sustainable commercial demand.

3.4. The State as Enabler and First Customer

At early stages, new production directions may face insufficient private demand, high costs of first series, and uncertainty of returns.

The state can play three roles:

1. **Builder of infrastructure** — invest in energy, networks, computing capacity, transport infrastructure, research, and necessary standards.
2. **First large customer** — form demand for solutions that the state needs regardless of commercial maturity: defence systems, critical infrastructure, energy equipment, robotics, and automation.
3. **Bearer of initial risk** — help pass the stage at which a technology already has practical value but has not yet reached the scale needed to attract sufficient private capital.

State orders should launch production and learning on real series, not become permanent compensation for inefficiency.

Defence procurement is especially important as one source of initial demand. However, it should not become the sole market for new technologies. Defence orders can serve as a start to scaling, not the limit of further development.

4. Dual-Use as a Scaling Mechanism

New technological platforms should be designed from the outset for both military and civilian applications.

For example, an autonomous unmanned platform may be used for:

- Reconnaissance.
- Logistics.
- Mapping.
- Agriculture.
- Infrastructure inspection.
- Energy monitoring.
- Search and rescue operations.

Common components remain: sensors + AI + autonomy + navigation + communications + power electronics + manufacturing.

The scaling mechanism:

> *State order → first series → learning curve → cost reduction → civilian applications → mass market → larger series → further cost reduction → export.*

In this way, public expenditure becomes investment in a future market rather than permanent support of a producer.

5. Why This Should Work

The new production function should look as follows:

> *Competitive energy → available computing → AI → robotics → automated production → high productivity → low unit cost → new competitive products → mass market → export.*

This differs fundamentally from permanent subsidisation.

A subsidy compensates for poor economics. Automation, energy, and new technologies change the economics of production itself.

6. Why the Solution Must Be European

Germany does not exist separately from the European economy. Its industry is linked to markets, suppliers, capital, infrastructure, and production competencies of other EU countries.

Therefore, the reconstruction of German industry inevitably affects the distribution of production and economic opportunities in Europe.

Other EU countries have their own rational interests:

- Obtain investment and jobs.
- Develop their own production competencies.
- Participate in new technological chains.
- Prevent excessive concentration of benefits from state support in one country.
- Preserve access to the single market.

- Reduce critical technological dependencies.
- Strengthen the European defence and industrial base.

If Germany builds a new production system primarily for its own advantage, other states may perceive it as an attempt to restore or strengthen German industrial concentration at their expense.

Then an economically successful programme may encounter political resistance capable of limiting its scale or derailing implementation.

But there is also the opposite risk: if Germany continues to lose its industrial base, the European system may be rebuilt without it as one of the main centres.

Therefore, the European dimension is necessary not for abandoning German interests, but for reconciling two tasks:

1. Preserve Germany's ability to independently shape industrial and technological policy.
2. Create conditions under which other EU countries obtain their own benefits from German industrial renewal.

7. How to Resolve the Contradiction with Other EU Countries

7.1. Principle: German Leadership + European Production

The programme should be built on the principle:

German technological leadership + distributed European production.

Germany should obtain a first-mover advantage: begin the transition earlier, accumulate experience, create technological platforms, develop production standards, and scale key competencies.

But this should not turn into an exclusive right of Germany to the entire industrial system being created.

Possible distribution of specialisations may include:

- Germany — industrial AI, robotics, machine tools, systems integration, and automation.
- France — energy technologies, aerospace, and defence systems.
- Italy — machinery and specialised manufacturing.
- Poland — production capacity, logistics, and defence-industrial directions.
- Central Europe — components and production chains.
- Northern Europe and Benelux — specialised technological, energy, and industrial competencies.

This is not a final allocation of roles and not a proposal to administratively assign specific industries to each country. The concrete configuration should emerge from existing advantages, demand, investment, and voluntary participation of companies.

What is essential is that European partners must be able to obtain real economic benefit from the new system.

7.2. From National Subsidies to European Production Chains

If state support leads only to concentration of production in Germany, other countries may demand similar mechanisms for their own companies or oppose the programme.

To reduce this risk, part of the support should be linked to the formation of production chains accessible to European partners.

This may include:

- Joint technological projects.
- Distribution of component production.
- Joint defence procurement.
- Compatible technical standards.
- Joint infrastructure projects.
- Access of companies from different countries to production and research programmes.
- Development of cross-border supplier chains.

At the same time, every job should not be artificially distributed among countries. Such distribution can destroy the economic effect of specialisation.

The goal is not to place all production evenly, but to create a system in which other countries have grounds to support the project because they participate in its economic results.

7.3. Europe as a Source of Scale

The German domestic market will not always be sufficient for the payback of new production platforms.

The European market allows:

- Increasing series.
- Distributing research costs.
- Using compatible standards.
- Expanding demand.
- Reducing average costs.
- Forming a stronger export position.

In this model, Germany acts not as the sole recipient of European resources, but as one of the key creators of production opportunities used by the wider European economy.

7.4. Political Exchange

Germany should not in advance abandon industrial advantage for the sake of formal balance.

But it must show that this advantage creates opportunities for other countries as well.

The political exchange looks as follows:

Germany obtains the opportunity to be the first to rebuild its production model and retain the role of one of the technological centres of Europe.

In return:

Other countries obtain access to new production chains, investment, technologies, markets, and joint projects.

This does not eliminate all conflicts of interest. But it changes the structure of the conflict: instead of opposition to German strengthening, the possibility of negotiating participation in the new production system appears.

8. Germany as One of the Centres of European Political Balance

Germany’s economic weight matters not only for its internal well-being.

It also affects the distribution of political opportunities within the EU.

In the European system, Germany and France have traditionally acted as two large centres capable of forming initiatives and agreeing on the directions of Union development. Their interests do not always coincide, but interaction between them constitutes an important part of European political dynamics.

If Germany’s industrial base weakens substantially, its ability to participate in shaping European policy will also change.

This may reduce the number of economically and politically independent centres capable of initiating large projects and gathering coalitions around them.

However, it does not follow that Germany’s weakening will inevitably lead to the collapse of the EU or to sole management by France or European institutions. These are possible political scenarios, not established consequences.

For the proposed programme, a narrower conclusion is important:

Preserving Germany as an independent industrial centre allows it to remain a participant in the formation of European decisions, rather than only adapting to decisions taken by others.

This serves German interests and at the same time can support political balance within the EU.

But this advantage should not be built on confrontation with France or Brussels. It must be realised through Germany’s ability to propose projects that create value for other participants.

The task is not to protect German dominance, but to preserve Germany’s ability to be one of the architects of European development.

9. Two Political Risks of European Restructuring

This is the central political problem of the entire initiative.

Germany must transform its economy, but the restructuring itself takes place inside the EU — a system in which other states have their own interests, mechanisms of influence, and ideas about the future distribution of industrial production.

Therefore, it is not enough to develop an economically effective programme. It is necessary to take into account who will control its implementation, who will receive the main benefits, and who will be able to determine the final architecture of the new European industry.

9.1. Risk of the First Kind: Germany Rebuilds While Trying to Preserve Its Previous Position

Suppose Germany understands the need for automation and transition to a new production model.

It restores energy competitiveness, invests in AI and robotics, supports new industries, uses state orders, and creates high-tech markets.

However, the political goal remains the same: to preserve for Germany the largest possible share of industrial production and its previous position as the main industrial centre of Europe.

In that case, other countries may perceive the programme not as European renewal, but as an attempt to use the transition for further concentration of economic power in Germany.

The following chain arises:

> *Germany modernises production → obtains an advantage in cost and technology → concentrates investment and new production → other countries fear strengthening of German superiority → a coalition forms against the German programme → European mechanisms of coordination and implementation of the project are blocked or limit it.*

Such resistance may come from different sources and for different reasons:

- France may insist on stronger pan-European coordination of industrial policy.
- Countries of Central and Eastern Europe may demand greater access to investment and production chains.
- Italy and other industrially developed countries may seek similar support for their enterprises.
- The European Commission may fear subsidies that violate competition and the balance of the single market.
- National governments may oppose concentration of benefits in one country.

These interests will not necessarily lead to a single coalition. But if the German programme is perceived as a project of national dominance, the probability of coordinated resistance may increase.

Result: Germany successfully develops a new production model but cannot fully implement it because of political conflict inside the EU.

This is a risk not of technological error, but of error in political positioning.

How to reduce it

Germany must separate two concepts:

- Leadership in creating and scaling new technologies.
- Exclusive right to the economic benefits of these technologies.

The first should remain the goal of the programme. The second should not be its mandatory condition.

Germany may strive for advantage in systems architecture, engineering, industrial equipment, automation, and high-margin components, while simultaneously allowing the development of distributed European production chains.

Thus, it retains a strategic advantage but reduces the probability that other countries will consider it necessary to unite to limit it.

9.2. Risk of the Second Kind: Europe Rebuilds Industry Without Germany as an Independent Centre

There is also the opposite scenario.

Other EU countries may be interested not only in restoring German industry, but also in changing the previous distribution of production opportunities in Europe.

Within such logic, the weakening of Germany may be perceived as an opportunity to redistribute production, capital, investment, and political influence among other European centres.

This does not mean that the EU has a single proven plan of deliberate weakening of Germany. European institutions and national governments have different interests, and the preservation of German industry itself creates significant benefits for the European economy.

Nevertheless, for the design of German strategy it is necessary to take into account a scenario in which the interests of individual European participants in the redistribution of production may come into conflict with Germany's interest in preserving its own position.

Suppose Germany continues to lose industrial competitiveness and postpones transformation.

Then the following processes may occur sequentially:

> *Weakening of German industry → decline in investment attractiveness of Germany → relocation of production and competencies → strengthening of alternative European centres → change in the distribution of bargaining power → formation of a new industrial architecture without Germany as an independent systemic leader.*

In such a situation, Germany may receive an offer to participate in the new European model, but already on terms formed by others.

It will retain individual important industries and competencies, but its role will be limited to certain functions within a broader system.

Instead of independently determining the directions of industrial development, Germany will find itself in the position of one of the participants in an already formed architecture.

This is fundamentally different from voluntary participation in the European division of labour.

The problem is not that Germany will produce only part of European output. The problem is that it may lose the ability to independently determine which production systems to create, where to direct investment, and how to develop its own industrial base.

In this scenario, formal membership in the EU and participation in joint programmes do not guarantee preservation of the previous degree of independence.

How to reduce it

Germany must begin transformation before the loss of its production base leads to a critical reduction of its bargaining opportunities.

It must retain its own competencies in the following areas:

- Development and integration of complex industrial systems.
- Industrial AI and robotics.
- Production equipment.
- Energy technologies.
- Key components and critical production processes.
- Research and the transition from development to serial production.

- The ability to independently finance and launch strategic projects.

Participation in the European industrial system should strengthen these capabilities, not replace them.

European integration should expand the scale of German industrial policy, not deprive Germany of the ability to conduct its own industrial policy.

9.3. Conclusion: Between the Two Risks There Is No Simple Compromise

The first risk arises when Germany tries to preserve its previous position by using the new production model primarily to strengthen its own industrial concentration.

The second arises when Germany weakens its own production base so much that it ceases to be an independent participant in the formation of the new European system.

Therefore, the correct transition does not amount to choosing a middle ground between national dominance and European redistribution.

A sequence of actions is necessary:

1. Germany begins transformation while it still has sufficient industrial, technological, and political resources.
2. It forms its own new production competencies and obtains a first-mover advantage.
3. The new production platform is designed from the outset with the possibility of European scaling.
4. Other countries obtain grounds to participate in the project because it creates real economic opportunities for them.
5. Germany retains the ability to independently initiate and develop strategic projects while participating in the pan-European system.

Thus, Germany should neither try to restore its previous position exclusively at the expense of other countries, nor agree to a role completely determined by other participants.

It must preserve the ability to shape its own future and make the realisation of this interest compatible with the development of Europe.

10. China: Competition Through Change of the Production Model

China is simultaneously a competitor, a supplier, a market, and a source of technological pressure.

The response should not be limited to tariffs or protection of individual industries.

Trade measures may be used where economic and legal grounds exist. But they do not replace productivity improvements.

The strategic task is to change the production function of European industry.

Conditionally:

> *China: scale of production + competitive cost of resources + automation + developed supplier chains.*

> *Europe: competitive energy + computing capacity + AI + robotics + engineering competencies + automated production.*

This does not mean that the listed advantages are already equally available to both sides. On the contrary, the programme must identify concrete technological and economic gaps and consistently reduce them.

The goal is to restore to European producers the ability to compete without permanent compensation for cost differences.

11. The United States: Make European Production Attractive

The United States may attract European investment, technologies, and production capacity through a combination of energy conditions, state incentives, access to capital, and a large domestic market.

Europe cannot sustainably solve this problem solely by prohibiting the relocation of production.

It is necessary to make locating production in Europe economically attractive.

This requires:

- Predictable energy policy.
- Available computing capacity.
- High productivity.
- Investment in infrastructure.
- Scalable production chains.
- Clear rules of state support.
- Access to a large internal market.

The United States may remain a partner in research, technologies, capital, and trade. At the same time, Europe must retain sufficient own production competencies so that cooperation does not turn into one-sided dependence.

12. Coalition of Implementation

The programme requires not only an economic model, but also a coalition of participants with different interests.

Potential participants:

- **Industry.** Obtains cost reduction, access to new technologies, and the ability to create competitive products.
- **Mittelstand.** Obtains the opportunity to automate production, participate in new supplier chains, and expand exports.
- **Workers and trade unions.** Interested in sustainable industrial employment, qualifications, and participation in productivity growth. At the same time, automation may lead to the reduction of individual professions, so programmes of retraining and transition between sectors are necessary.
- **Energy sector.** Obtains long-term demand, but must adapt to changes in the structure of generation and consumption.
- **Federal government and Länder.** Obtain a basis for industrial growth, tax base, and investment.

- **Defence sector.** Obtains the opportunity to scale production and develop dual-use technological platforms.
- **AI and computing infrastructure sector.** Obtains demand for computing, automation, and integration of technologies into physical production.
- **Other EU countries.** Obtain opportunities for participation in production chains, joint orders, and technological projects.
- **Consumers.** May obtain more competitive products and new services, although the distribution of benefits will depend on concrete markets and results of the programme.

The coalition should not be built on the assumption that the interests of all participants coincide. Its task is to create a construction in which cooperation gives each sufficiently concrete benefits to support the transition.

13. Resistance and Ways to Account for It

The programme will encounter several types of resistance.

Existing industries interested in preserving the previous model

They may demand permanent compensation of costs instead of restructuring production processes.

Response: link support to measurable results, productivity growth, and transition to new markets.

Companies that prefer protection of competition to investment

Some enterprises may count on import restrictions and subsidies without changing their own production model.

Response: use support for modernisation and creation of new competencies, not for indefinite preservation of uncompetitive processes.

EU states fearing concentration of production in Germany

They may oppose the programme if its benefits are distributed mainly inside Germany.

Response: provide for European production chains, joint projects, and access to new markets, while preserving economic incentives for specialisation.

European institutions fearing a subsidy race

They may view national programmes as a threat to the single market.

Response: where possible, use transparent support criteria, compatible standards, and pan-European mechanisms where they increase scale and reduce political conflicts.

Elements of the existing energy system

Participants who have invested capital in current infrastructure may be interested in its maximum utilisation and prolongation of the previous model.

Response: evaluate infrastructure by its future systemic value, including reserve and resilience, and not only by the volume of current use.

Political participants interested in changing the distribution of industrial influence in Europe

They may support pan-European restructuring but strive for a configuration in which Germany loses independence in determining industrial policy.

Response: participate in the formation of the European system while simultaneously retaining own technological competencies, production capabilities, and the ability to initiate strategic projects.

None of these responses guarantees the elimination of resistance. Their task is to take conflicts of interest into account in advance and reduce the probability that they will block an economically necessary transition.

14. The State Must Have an Exit Condition

The programme should not create permanent dependence of industry on state financing.

Implementation is proposed to be divided into three phases.

Phase I. Creation of Capabilities

Approximately the first 0–3 years, with adjustment of timelines by direction.

- Development of energy and network infrastructure.
- Increase in availability of computing capacity.
- Launch of research and production projects.
- Formation of first state orders.
- Development of standards.
- Creation of first series of new products.
- Development of cooperation between industry, research centres, and suppliers.

Phase II. Scaling

Approximately 3–7 years.

- Expansion of production.
- Increase in private investment.
- Development of civilian markets.
- Inclusion of companies from other EU countries.
- Cost reduction.
- Growth of exports.
- Transition from experimental projects to sustainable production platforms.

Phase III. Reduction of Direct Support

Approximately 7–10 years.

Support for concrete products and companies should be reduced as the following conditions are achieved:

- The product is competitive without permanent cost compensation.
- A sustainable civilian or export market has formed.
- Private investment has become the main source of expansion.
- Production has reached sufficient scale.
- The enterprise is able to develop without state orders for survival.

Timelines are guidelines, not a guaranteed schedule. Different directions will pass these phases at different speeds.

If a direction does not achieve the stated goals, its design must be revised, executors changed, or support terminated.

The state must help create competitive production, not become a permanent source of demand for products that the market does not need.

15. What the New Industrial System Should Look Like

The proposed architecture combines energy, technological, production, and market elements.

> *Energy → computing infrastructure → AI → robotics → automated production → new high-tech products → civilian markets and state demand → European scaling → export → investment in the next generation of technologies.*

The state accelerates the transition through infrastructure, initial demand, and acceptance of part of the early risk.

The private sector ensures commercial realisation, scaling, and development of competitive products.

The European market allows increasing the scale of production and distributing technological and production competencies among several centres.

As a result, Germany should move from a model based mainly on the export of traditional industrial products to a model in which its advantages are also determined by:

- Productivity.
- Automation.
- Industrial software.
- Robotics.
- Integration of complex systems.
- Engineering architecture.
- Ability to quickly create new products and scale them.

16. What New Political-Economic Reality the Programme Creates

If the programme is implemented, its result should be not simply a modernised German industry.

It can create a different configuration of economic relations inside Europe.

Germany

Germany retains a substantial industrial base, but its advantage gradually shifts from the previous structure of production to technological platforms, automation, industrial equipment, and integration of complex systems.

Its role in Europe is based not only on the size of the economy, but also on the ability to initiate new production directions.

Other EU countries

They obtain additional opportunities to participate in European value chains.

At the same time, their interests will not necessarily coincide with those of Germany. Disagreements over the distribution of investment, subsidies, and production will remain.

But the presence of mutual economic benefit creates a basis for negotiations instead of automatic opposition to German industrial renewal.

European economy

It obtains the opportunity to move from a collection of national programmes to more interconnected production platforms capable of competing on large external markets.

This does not require a complete abandonment of national industrial policy. Rather, national programmes should be organised so that their compatibility increases the scale and resilience of the European economy.

New and old dependencies

The programme may reduce dependence on:

- Individual traditional export markets.
- Permanent compensation of costs of old production.
- Excessive role of cheap human labour.
- Individual external suppliers of critical technologies.
- One configuration of the energy system.

At the same time, the importance will grow of:

- Available and reliable energy.
- Computing capacity.
- Semiconductors and critical components.
- Qualified specialists.
- Compatible European standards.
- Ability to maintain complex production platforms.

Strategic autonomy does not mean the absence of external dependencies. It means the ability to diversify them, manage them, and retain critical competencies inside one’s own economic system.

17. Checking the Result After Five Years

After five years it is necessary to evaluate not the number of announced programmes, agreements, and subsidies, but the change in real production capabilities.

Key questions:

1. Has energy uncertainty for industry decreased?
2. Have investments in automation and productivity increased?
3. Have new products capable of competing without permanent cost compensation appeared?
4. Have private investment and serial production increased?
5. Have new civilian markets emerged for technologies that initially developed through state demand?
6. Have key engineering and production competencies been retained in Germany?
7. Has the export of new high-tech products increased?
8. Have European production chains become deeper and larger in scale?

9. Does Germany retain the ability to independently initiate and implement large industrial projects?
10. Has the programme avoided becoming a permanent transfer of funds to incumbent businesses without substantial changes in the production model?

If output grows while Germany loses the ability to shape the technological and industrial architecture, the distribution of strategic capabilities must be examined separately.

If technological development succeeds but fails to progress to serial manufacturing, the conditions for scaling must be reconsidered.

If public procurement increases but civilian markets fail to emerge, the commercial potential of the relevant areas should be reassessed.

18. Final Logic

The problem

Expensive energy and changes in global competition → loss of price competitiveness → declining production and investment → erosion of the industrial base → declining economic and political weight for Germany.

The future challenge

More affordable energy, computing, and robotics may change production economics in favour of countries that adopt automation and scaling earlier.

The solution

Competitive energy + AI and computing capacity + robotics + automated manufacturing + new high-tech markets + initial public demand + dual-use design + European scaling.

First political risk

Germany uses modernisation to preserve its previous concentration of industrial power → other countries fear German dominance → a coalition forms against the programme.

Second political risk

Germany’s industrial base weakens too far → a new European system is built around other centres → Germany accepts an assigned role and loses the ability to determine its own industrial development.

Adaptation

Germany must transform its economy while it still has sufficient capacity to influence the architecture of the transition. At the same time, the new production platform must support European scaling from the outset, and participation must generate tangible economic benefits for other countries.

Condition for success

European integration must expand Germany’s capabilities rather than replace them. German industrial renewal must create value for Europe without depriving Germany of its ability to pursue an independent industrial strategy.

Conclusion

Germany faces more than a problem of competitiveness in individual industries. It faces a question about the role it will play in the next generation of European industry.

An attempt to preserve the existing structure without changing the economics of production may accelerate the loss of industrial position.

However, simply accepting a new European model formed while Germany is weakening does not guarantee that Germany will retain the ability to determine its own industrial development.

The transformation must therefore begin before industrial contraction severely restricts Germany’s options.

Germany should use its existing engineering capabilities, manufacturing infrastructure, and political weight to transition to a new production model built on competitive energy, AI, robotics, automation, and new markets.

At the same time, this model must be compatible with European scale, giving other countries tangible economic benefits from participation.

The objective is neither to preserve German dominance at any cost nor to abandon German interests in the name of European balance.

It is to preserve Germany’s ability to shape its own industrial future while creating a European system in which doing so becomes a source of shared economic strength.

> *Germany cannot preserve the old system merely to preserve its position.*
> *But neither can it afford to lose its position in the process of building a new system.*
> *It must use its remaining industrial and political weight to transition to the new system—and make that system European.*